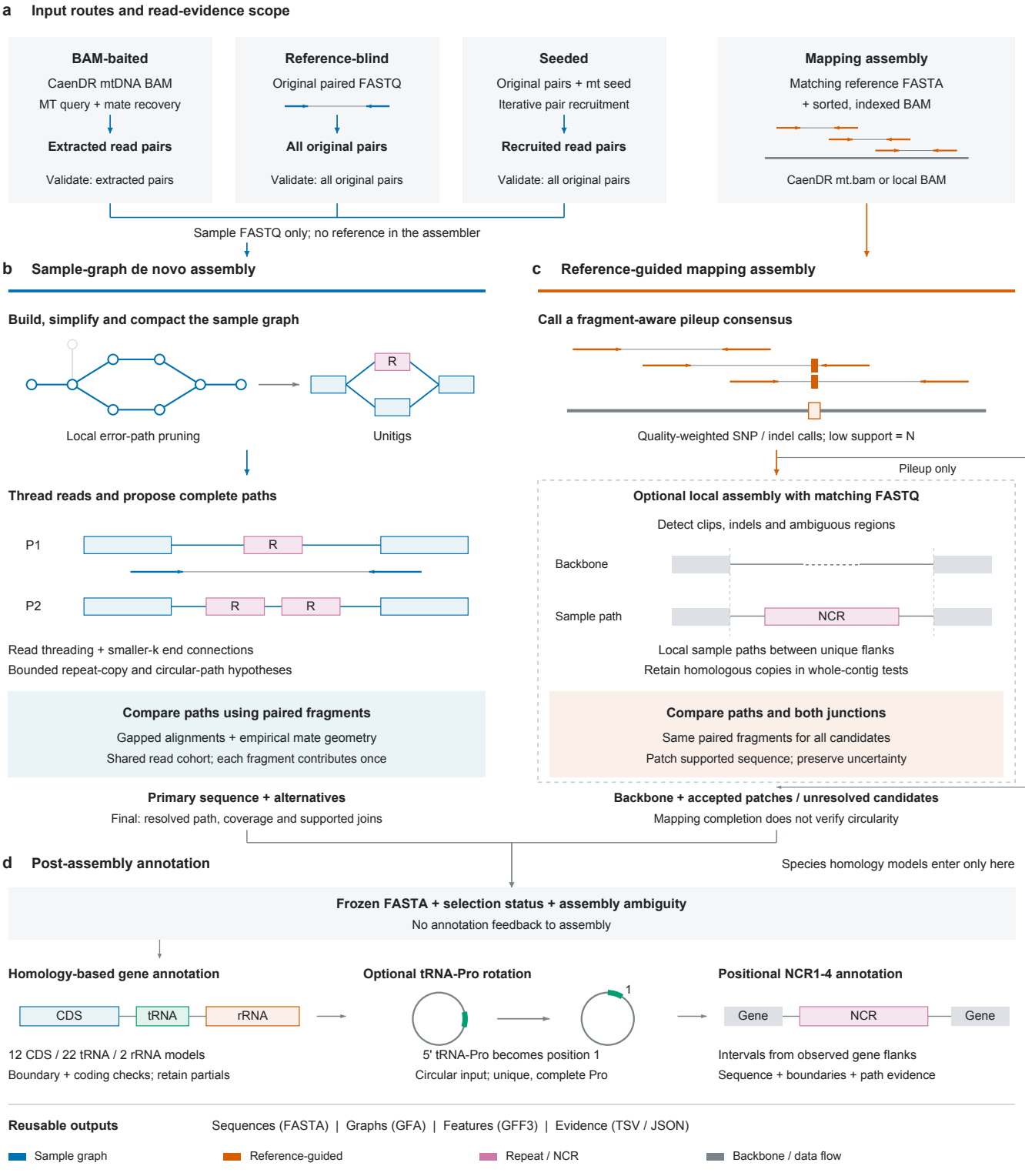

a
Input routes and read-evidence scope
BAM-baited
CaenDR mtDNA BAM
MT query + mate recovery
Extracted read pairs
Validate: extracted pairs
Reference-blind
Original paired FASTQ
All original pairs
Validate: all original pairs
Seeded
Original pairs + mt seed
Iterative pair recruitment
Recruited read pairs
Validate: all original pairs
Mapping assembly
Matching reference FASTA
+ sorted, indexed BAM
CaenDR mt.bam or local BAM
Sample FASTQ only; no reference in the assembler
b
Sample-graph de novo assembly
Build, simplify and compact the sample graph
R
Local error-path pruning
Unitigs
Thread reads and propose complete paths
P1
R
P2
R
R
Read threading + smaller-k end connections
Bounded repeat-copy and circular-path hypotheses
Compare paths using paired fragments
Gapped alignments + empirical mate geometry
Shared read cohort; each fragment contributes once
Primary sequence + alternatives
Final: resolved path, coverage and supported joins
c
Reference-guided mapping assembly
Call a fragment-aware pileup consensus
Quality-weighted SNP / indel calls; low support = N
Pileup only
Optional local assembly with matching FASTQ
Detect clips, indels and ambiguous regions
Backbone
Sample path
NCR
Local sample paths between unique flanks
Retain homologous copies in whole-contig tests
Compare paths and both junctions
Same paired fragments for all candidates
Patch supported sequence; preserve uncertainty
Backbone + accepted patches / unresolved candidates
Mapping completion does not verify circularity
d
Post-assembly annotation
Species homology models enter only here
Frozen FASTA + selection status + assembly ambiguity
No annotation feedback to assembly
Homology-based gene annotation
CDS
tRNA
rRNA
12 CDS / 22 tRNA / 2 rRNA models
Boundary + coding checks; retain partials
Optional tRNA-Pro rotation
1
5' tRNA-Pro becomes position 1
Circular input; unique, complete Pro
Positional NCR1-4 annotation
Gene
NCR
Gene
Intervals from observed gene flanks
Sequence + boundaries + path evidence
Reusable outputs
Sequences (FASTA) | Graphs (GFA) | Features (GFF3) | Evidence (TSV / JSON)
Sample graph
Reference-guided
Repeat / NCR
Backbone / data flow